

Name: _____

Teacher: _____

SYDNEY TECHNICAL HIGH SCHOOL



Year 11 Mathematics Extension 1

Preliminary Assessment Task 1

May 2012

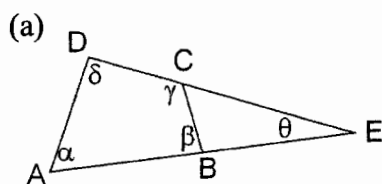
Time allowed: 70 minutes

Instructions: Please take notice and act upon ALL of these.

- Show full working.
- Start each question on a new page.
- Full marks may not be awarded for careless or badly arranged work.
- Non-programmable calculators may be used.
- This paper must be handed in on top of your answer booklets.
- Answers must be written in blue or black pen.
- Answers must be arranged in order and stapled securely.

Question 1**9 marks**

- (a) Factorise fully $x(x-y)^2 - xz^2$. 2
- (b) Simplify $\frac{\frac{x+y}{y-x}}{\frac{x-y}{y-x}}$. 2
- (c) Factorise $64 - 27k^3$. 1
- (d) Solve $\frac{x}{3} - 2 < \frac{x}{2} - 3$. 1
- (e) Find the points of intersection of $x^2 + y^2 = 16$ and $x - y = 2$. 3

Question 2 (Start a new page)**10 marks**

In quadrilateral ABCD there are no parallel sides. 3

Use triangle and other geometric properties to show that the interior angle sum of such a quadrilateral is 360° .

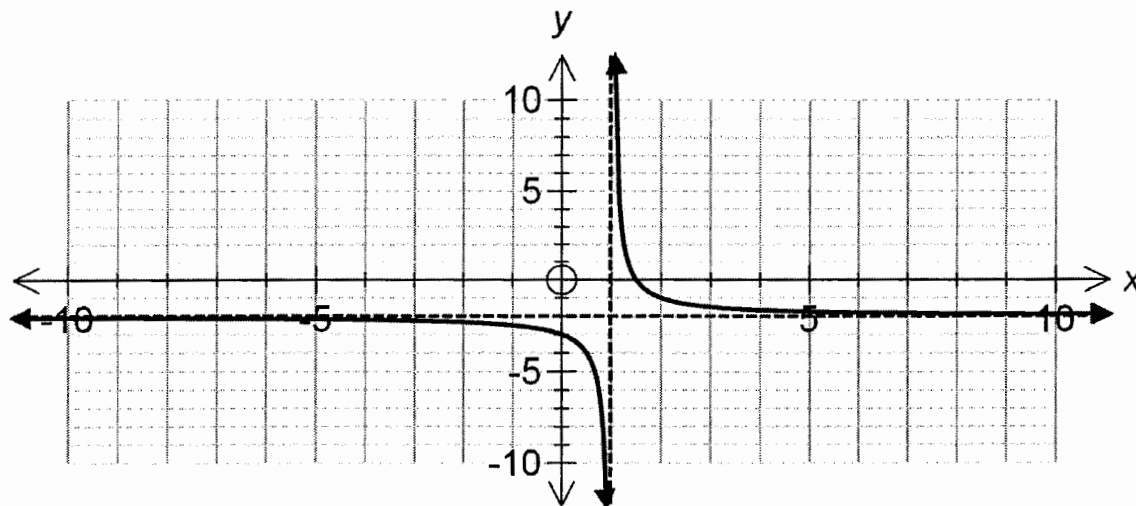
Show reasoning.

- (b) For $y = \sqrt{x-2}$:
- (i) Is it a function? (Show reasoning) 1
- (ii) State its domain and range. 2
- (iii) Sketch the curve showing any significant features. 2
- (c) (i) Factorise $x^2 - x - 6$. 1
- (ii) Hence sketch $y = |x^2 - x - 6|$. 2

Question 3 (Start a new page)

9 marks

- (a) Find rational numbers a and b such that $a + b\sqrt{5} = \frac{\sqrt{5}}{3 + \sqrt{5}}$. 2
- (b) The curve shown is a translation of $y = \frac{1}{x}$. Write its equation. 2



- (c) Show whether the function $f(x) = \frac{8x}{x^2 + 9}$ is odd, even or neither. 2
- (d) Solve the inequality and graph the solution on a number line. 3
- $$\frac{3}{x-1} < \frac{5}{2}$$

Question 4 (Start a new page)

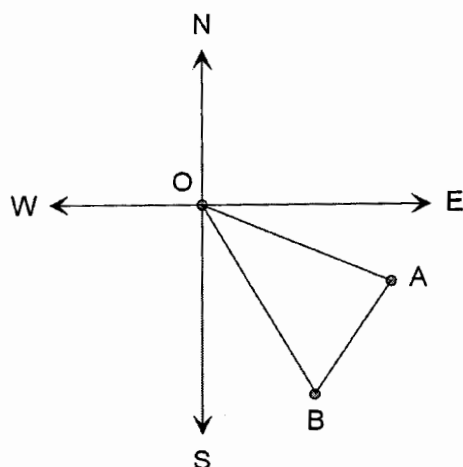
10 marks

- (a) Solve $2\sin\theta = \sqrt{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 2
- (b) Solve $\cos 2\theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 3
- (c) Find the exact value of $\sec 315^\circ$. 2
- (d) Prove the identity $(1 - \cos \theta)(1 + \sec \theta) = \sin \theta \tan \theta$. 3

Question 5 (Start a new page)

9 marks

- (a) If $\tan\theta = -\frac{3}{4}$ and θ is obtuse, find $\sin\theta$ and $\cos\theta$. 3
- (b) Eliminate θ from the set of equations $x = a\sec\theta$ and $y = b\tan\theta$. 2
- (c) A ship sails from O to a point A 60 km on a bearing of 125° . It then changes to a bearing of 200° and sails to a point B , which has a bearing from O of 150° .

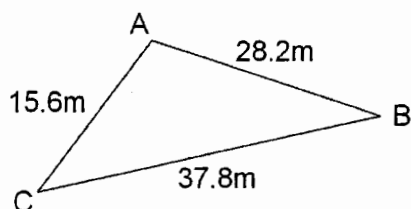


- (i) Copy the diagram into your workbook and find the values of the internal angles of the triangle OAB . 2
- (ii) What is the distance of B from O ? 2

Question 6 (Start a new page)

10 marks

- (a) Find the largest angle in the triangle (to the nearest minute). 2



- (b) (i) Sketch $y = 2 - x^2$. 1
- (ii) Hence or otherwise, solve $|x| + x^2 \geq 2$ 3
- (c) (i) Find any vertical asymptotes of the curve $y = \frac{x^2 - 3}{x^2 - 4}$. 1
- (ii) Sketch the curve, including any vertical and horizontal asymptotes. 3

End of Paper

SOLUTIONS PRELIMINARY EXTENSION 1 ASSESSMENT 1

Q1 a) $x(x-y)^2 - xz^2 = x[(x-y)^2 - z^2]$
 $= x(x-y-z)(x-y+z)$

b) $\frac{x+y}{\frac{y}{x}} = \frac{x^2+y^2}{xy}$
 $\frac{x}{y} - \frac{y}{x} = \frac{x^2-y^2}{xy}$

c) $64 - 27k^3 = 4^3 - (3k)^3 = \frac{x^2+y^2}{x^2-y^2}$
 $= (4-3k)(16+12k+9k^2)$

d) $\frac{x-2}{3} < \frac{x-3}{2}$

$\therefore 2x-12 < 3x-18$

$6 < x$

$\therefore x > 6$

e) $\left. \begin{aligned} x^2+y^2 &= 16 \\ x-y &= 2 \end{aligned} \right\}$

$y^2 = (2-x)^2$

$\therefore x^2 + (2-x)^2 = 16$

$\therefore x^2 + 4 - 4x + x^2 = 16$

$\therefore 2x^2 - 4x - 12 = 0$

$\therefore x^2 - 2x - 6 = 0$

$\therefore x = \frac{2 \pm \sqrt{4 - 4 \times 1 \times -6}}{2}$

$= 2 \pm \sqrt{28}$

$\therefore x = 1 \pm \sqrt{7}$

$\therefore y = -1 \pm \sqrt{7}$

Q2

a) In $\triangle ADE$, $\theta + \alpha + \delta = 180$
 $\therefore \theta = 180 - (\alpha + \delta)$

In $\triangle CBE$, $\angle ECB = 180 - \alpha$
 $\therefore \angle ECB = 180 - \alpha$

and $\angle EBC = 180 - \beta$

$\therefore \theta = 180 - ((180 - \alpha) + (180 - \beta))$

$= \alpha - 180 + \beta$

$\therefore 180 - (\alpha + \delta) = \alpha - 180 + \beta$

$\therefore \alpha + \delta + \alpha + \beta = 360^\circ$

$\therefore$ the interior anglesum of a quadrilateral with no parallel sides is 360° .

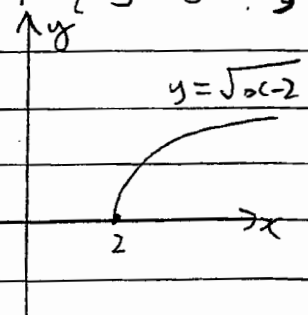
b) (i) $y = \sqrt{x-2}$

Yes, it is a function as the convention is to take the positive root and so for each x value there is at most one y value.

(ii) domain $\{x : x \geq 2\}$

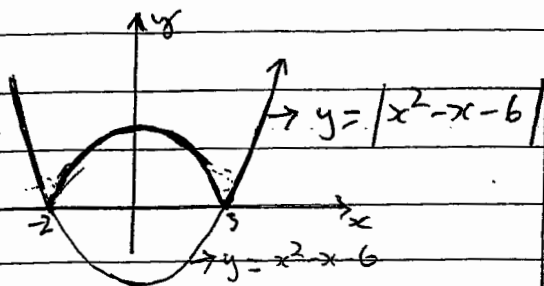
range $\{y : y \geq 0\}$

(iii)



Q) (i) $x^2 - x - 6 = (x-3)(x+2)$

(ii)



3 a) $9 + b\sqrt{5} = \frac{\sqrt{5}}{3+\sqrt{5}} \times \frac{3-\sqrt{5}}{3-\sqrt{5}}$

$$= \frac{3\sqrt{5} - 5}{9 - 5}$$

$$= -\frac{5}{4} + \frac{3\sqrt{5}}{4}$$

$$\therefore a = -\frac{5}{4} \text{ and } b = \frac{3}{4}$$

d) $6(x-1) < 5(x-1)^2$
 $6x - 6 < 5x^2 - 10x + 5$

$$5x^2 - 16x + 11 > 0$$

Let $(5x-11)(x-1) = 0$

$$\therefore x = 1, 2\frac{1}{5}$$

x	0	2	3
	✓	x	✓

$$\therefore x < 1, x > 2\frac{1}{5}$$

$$\begin{aligned} & \frac{(-x)^2 + 9}{x^2 + 9} \\ &= \frac{-8x}{x^2 + 9} \\ &= -f(x) \end{aligned}$$

$\therefore f(x)$ is odd

c) $f(x) = \frac{8x}{x^2 + 9}$

b) $y = \frac{1}{x-1} - 2$

Now, $f(-x) = \frac{-8x}{(-x)^2 + 9}$ or $y + 2 = \frac{1}{x-1}$

$$= \frac{-8x}{x^2 + 9}$$

$$= -f(x)$$

$\therefore f(x)$ is odd

Q4 a) $2\sin\theta = \sqrt{2}$ for $0^\circ \leq \theta \leq 360^\circ$

$$\therefore \sin\theta = \frac{\sqrt{2}}{2}$$

$$= \frac{1}{\sqrt{2}}$$

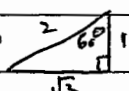


$$\therefore \theta = 45^\circ, 135^\circ$$

b) $\cos 2\theta = \frac{1}{2}$ for $0^\circ \leq \theta \leq 360^\circ$

$$\therefore 2\theta = 60^\circ, 300^\circ, 420^\circ, 660^\circ$$

$$\therefore \theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$$



d) LHS = $(1 - \cos\theta)(1 + \sec\theta)$

$$= 1 + \sec\theta - \cos\theta - 1$$

$$= \sec\theta - \cos\theta$$

$$= \frac{1}{\cos\theta} - \cos\theta$$

$$= \frac{1 - \cos^2\theta}{\cos\theta}$$

$$= \frac{\sin^2\theta}{\cos\theta}$$

$$= \sin\theta \tan\theta$$

$$= \text{RHS} \quad \text{A.E.D.}$$

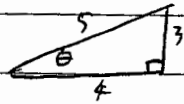
e) $\sec 315^\circ = \frac{1}{\cos 315^\circ}$

$$= \frac{1}{\cos 45^\circ}$$

$$= \frac{1}{\frac{1}{\sqrt{2}}}$$

$$= \sqrt{2}$$

Q5 a) $\tan \theta = -\frac{3}{4}$ and θ is obtuse.



θ in 2nd quadrant

$$\therefore \sin \theta > 0 \text{ and } \cos \theta < 0$$

$$\therefore \sin \theta = \frac{3}{5} \text{ and } \cos \theta = -\frac{4}{5}$$

$$\left. \begin{aligned} x &= a \sec \theta \\ y &= b \tan \theta \end{aligned} \right\}$$

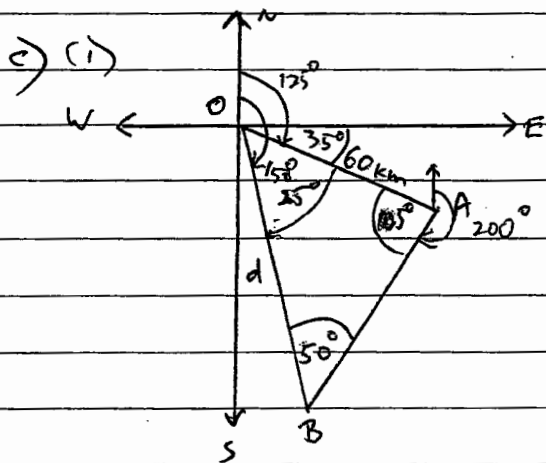
$$x^2 = a^2 \sec^2 \theta \rightarrow \sec^2 \theta = \frac{x^2}{a^2}$$

$$y^2 = b^2 \tan^2 \theta \rightarrow \tan^2 \theta = \frac{y^2}{b^2}$$

$$\text{Now, } \sec^2 \theta = 1 + \tan^2 \theta$$

$$\therefore \frac{x^2}{a^2} = 1 + \frac{y^2}{b^2}$$

$$\therefore \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \quad \text{Q.E.D.}$$



(ii) $\frac{d}{\sin 105^\circ} = \frac{60}{\sin 50^\circ}$

$$\therefore d = \frac{60 \sin 105^\circ}{\sin 50^\circ}$$

$$\approx 75.66$$

$$\approx 76 \text{ km to nearest km.}$$

Q6

$$a) \cos A = \frac{15.6^2 + 28.2^2 - 37.8^2}{2 \times 15.6 \times 28.2}$$

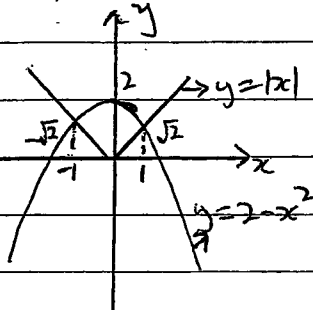
$$= -0.4435351887$$

$$\therefore A = 116.3296586$$

$$\approx 116^\circ 20'$$

A is opposite the longest side.
 $\therefore$ It is the largest angle.

$$b) (i) y = 2 - x^2$$



Only the parabola needed as
 answer to part (i)

$$(ii) |x| + x^2 \geq 2$$

$$\Rightarrow |x| \leq 2 - x^2$$

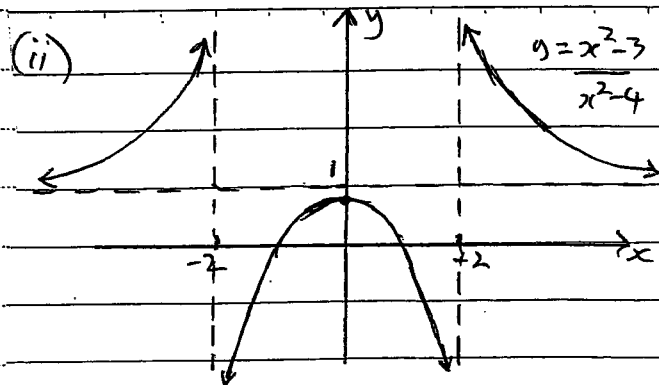
from the graph $x \leq -1$ or $x \geq 1$

$$c) (i) y = \frac{x^2 - 3}{x^2 - 4}$$

vertical asymptotes when $x^2 - 4 \neq 0$

$$\therefore (x - 2)(x + 2) = 0$$

$$\therefore x = \pm 2$$



$$\text{As } x \rightarrow -2^-, y \rightarrow \infty$$

$$\text{As } x \rightarrow -2^+, y \rightarrow -\infty$$

$$\text{As } x \rightarrow 2^-, y \rightarrow -\infty$$

$$\text{As } x \rightarrow 2^+, y \rightarrow \infty$$

$$\text{Now } y = \frac{x^2 - 3}{x^2 - 4}$$

$$= \frac{1 - \frac{3}{x^2}}{1 - \frac{4}{x^2}}$$

$$\text{As } x \rightarrow \infty, y \rightarrow 1$$

$$\text{As } x \rightarrow -\infty, y \rightarrow 1$$